

Penalized Trend Estimation from First Principles: A Step-by-Step Companion

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Abstract

This companion document explains the mathematical and computational ideas used in the Trend Estimation library for a reader who has not previously studied penalized smoothing. It develops finite differences, quadratic penalties, closed-form trend estimation, temporal forecasting, analytic sensitivity, and numerical smoothness selection. The active research paper is separate; this document is pedagogical and must remain consistent with the tested library and the derivations recorded under `notes/`.

1 The pure penalized smoother

Let $y = (y_1, \dots, y_n)^\top \in \mathbb{R}^n$ and let D_d compute finite differences of order d . We define

$$Q = D_d^\top D_d.$$

The pure penalized trend solves

$$\min_t \|y - t\|_2^2 + \lambda \|D_d t\|_2^2,$$

so

$$\widehat{t}_\lambda = (I + \lambda Q)^{-1} y.$$

The implementation solves the corresponding linear/spectral system rather than forming an explicit dense inverse.

2 Sensitivity to the penalty

Writing

$$S_\lambda = (I + \lambda Q)^{-1},$$

the inverse-matrix derivative gives

$$S'_\lambda = -S_\lambda Q S_\lambda.$$

Therefore

$$\widehat{t}'_\lambda = -S_\lambda Q \widehat{t}_\lambda,$$

and

$$\widehat{t}''_\lambda = 2S_\lambda Q S_\lambda Q \widehat{t}_\lambda.$$

3 Forecast loss

At a forecast origin T , no future observation may influence the fitted trend. Let H denote the continuation operator and let

$$r_T(\lambda) = y_{T+1:T+h} - HS_\lambda y_{\text{past}}.$$

Then

$$r'_T = HS_\lambda QS_\lambda y_{\text{past}},$$

and for

$$f_T(\lambda) = \frac{1}{h} r_T^\top r_T,$$

we obtain

$$f'_T(\lambda) = \frac{2}{h} r_T^\top HS_\lambda QS_\lambda y_{\text{past}}.$$

A second derivative is available as well and is derived step by step in `notes/derivative.md`.

4 Why optimize log-lambda?

Set

$$\theta = \log \lambda, \quad \lambda = e^\theta.$$

If $g(\theta) = f(e^\theta)$, then

$$g'(\theta) = \lambda f'(\lambda), \quad g''(\theta) = \lambda f'(\lambda) + \lambda^2 f''(\lambda).$$

This preserves positivity and makes multiplicative penalty scales easier to search.

5 Multiple stationary points

A validation objective need not be unimodal. The active numerical strategy is therefore to use a coarse log-penalty grid to locate derivative sign changes, apply a bracketed root solver to each interval, classify the stationary points, and compare all local minima with the search boundaries. Newton and direct bounded minimization remain useful comparison methods.

6 Chronological validation

Rolling-origin evaluation repeats

$$1:T_j \longrightarrow T_j+1:T_j+h$$

at several origins. This separates genuine forecasting from interpolation or trend reconstruction with future observations available on both sides.

7 Implementation map

Reusable mathematics belongs in `src/trend_estimation`. Internal derivations and research decisions live in `notes/`. The active research manuscript lives in `paper_forecast-optimal-smoothing/`.